

1. INPUT MODULE

α -Synuclein Pathology Data + Brain Connectome G



2. DATA & GRAPH PREPROCESSING

Kinetic Traces ($T = \{1, 3, 6\}$) $\longrightarrow$ Temporal Aggregation (z, Δ)

Connectome $G \longrightarrow$ Structural Edges & Vulnerability Features

3. DUAL-BRANCH ARCHITECTURE

Temporal Branch

GRU Sequence Modeling

$$z^{(T)} \in \mathbb{R}^{96}$$

Spatial Branch

GATv2 Contextual Attention

$$z^{(S)} \in \mathbb{R}^{96}$$



4. FEATURE FUSION

Concatenate $z^{(T)} \parallel z^{(S)} \longrightarrow$ 2-Layer MLP



5. PREDICTION & VALIDATION

Sigmoid $\longrightarrow$ Regional Infiltration Probability $\hat{y}_i \in [0, 1]$

Optimization $\longrightarrow$ BCE Loss + AdamW Updates